SCHOOL OF COMPUTER SCIENCE AND ENGINEERING · IILM UNIVERSITY

Introduction to Data Science

Internship / Training Report

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Roll Number: 2410031099 | Batch: 2024–2028
B.Tech CSE · Submitted to School of Computer Science and Engineering

Introduction

Purpose and framing of this report

Data science is an interdisciplinary area concerned with working with data to obtain useful information, identify patterns, support decisions, and communicate findings.

This report documents the learning theme, objectives, scope, methodology, tools, and expected outcomes of the certified activity "Introduction to Data Science."

It treats the activity as a structured learning and training project, rather than claiming industrial deliverables not stated on the certificate.

AT A GLANCE

COURSE

Introduction to Data Science

PLATFORM

Networking Academy

PROGRAM

Cisco Networking Academy

STUDENT

Satyam Kumar (2410031099)

COMPLETED

20 June 2026

Learning Platform & Problem Statement

Organization / Learning Platform

The certificate identifies Networking Academy as the provider, delivered through the Cisco Networking Academy program.

The certificate does not provide a detailed organization profile, course duration, module list, instructor details, or grading scheme.

Accordingly, this report does not attribute unsupported details to the certificate.

Problem Statement

Modern organizations generate and collect large quantities of data. A learner entering computer science needs a foundational understanding of how data is viewed, prepared, analyzed, interpreted, and communicated.

This report addresses the development of a structured introductory understanding of data science and its role in transforming raw data into meaningful information.

The focus is educational: building conceptual familiarity with the data science workflow and associated tools and practices.

Project Objectives

Six goals guiding the learning activity

Develop a foundational understanding of data science and its applications.

Understand the role of data in analytical and decision-making processes.

Become familiar with the major stages of a typical data science workflow.

Understand why data preparation and organization matter before analysis.

Build awareness of analytical thinking, patterns, trends, and interpretation.

Strengthen readiness for further study in data analytics and machine learning.

Scope of the Project

What this report covers — and what it deliberately does not claim

IN SCOPE

- The conceptual workflow of data science
- The role of data preparation and analysis
- Commonly used categories of tools
- Expected learning outcomes of the activity

NOT CLAIMED

- A specific industrial dataset
- A deployed model or production application
- An organization-specific business problem
- Details not present in the supplied certificate

Technologies & Tools — Conceptual Categories

The certificate confirms the course and platform but does not list specific software, languages, or datasets

Data

Raw material for analysis and interpretation

General concept

Data Preparation

Cleaning, organizing, and preparing data for analysis

General concept

Analysis

Finding patterns, relationships, and trends

General concept

Visualization

Communicating analytical findings clearly

General concept

Programming / Analytics Tools

Can support data manipulation and analysis

Not listed on certificate

Conceptual System Architecture

A generic introductory data-science flow — not a proprietary Cisco architecture

1. **Data Sources**

   Obtained from relevant structured or unstructured sources.

2. **Data Preparation**

   Organized, cleaned, and checked for consistency.

3. **Exploratory Analysis**

   Patterns, distributions, relationships, and trends examined.

4. **Model / Analytical Layer**

   Appropriate analytical methods applied to answer the problem.

5. **Interpretation**

   Results converted into meaningful conclusions.

6. **Communication**

   Findings presented through summaries, tables, or visuals.

Methodology

A progressive, seven-step learning workflow

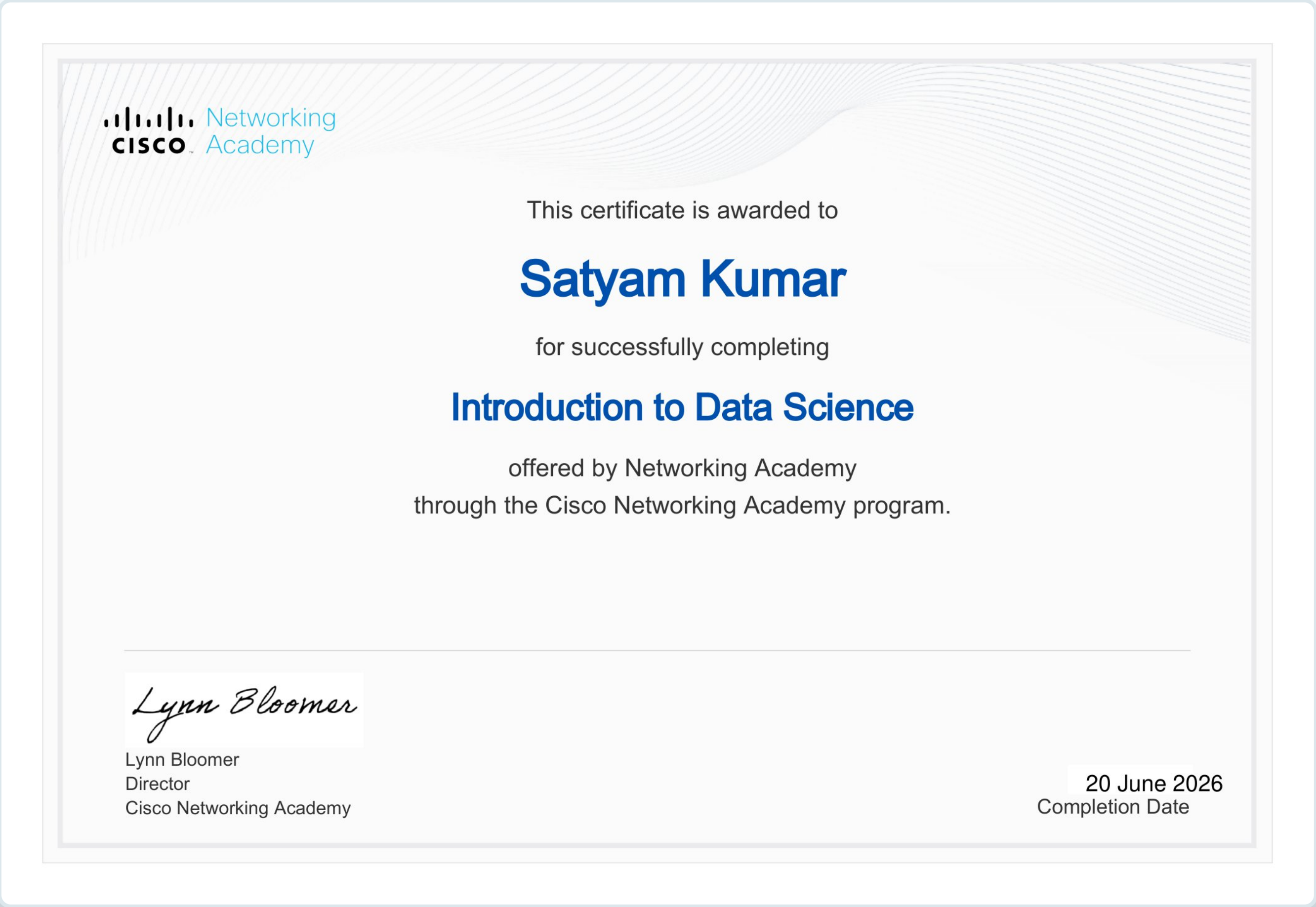
1. Understand the problem and define the information needed.
2. Identify the relevant data and understand its structure.
3. Prepare and organize the data for reliable analysis.
4. Explore the data to identify patterns, trends, and relationships.
5. Apply suitable analytical approaches where required.
6. Interpret the results and communicate the findings clearly.
7. Review the outcome and identify opportunities for further analysis.

Expected Outcomes

What the learner takes away from this training

- Improved understanding of the purpose and scope of data science.
- Awareness of the importance of data quality and preparation.
- Ability to describe a basic data science workflow.
- Improved analytical thinking and interpretation of findings.
- Better understanding of how data supports evidence-based decisions.
- Foundation for advanced learning in analytics, ML, and AI.

Certificate & Completion Evidence


Networking Academy

This certificate is awarded to

Satyam Kumar

for successfully completing

Introduction to Data Science

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer
Lynn Bloomer
Director
Cisco Networking Academy

20 June 2026
Completion Date

Issued to Satyam Kumar for successfully completing “Introduction to Data Science” through Networking Academy under the Cisco Networking Academy program. Signed by Lynn Bloomer, Director, Cisco Networking Academy. Completion date: 20 June 2026.

Conclusion

The “Introduction to Data Science” learning activity provided a foundation for understanding how data can be organized, analyzed, interpreted, and communicated to generate useful information.

This report has documented the purpose, objectives, scope, conceptual architecture, methodology, and expected learning outcomes of the activity.

Successful completion is evidenced by the certificate issued through Networking Academy under the Cisco Networking Academy program on 20 June 2026 — a foundation for continued study in data analytics, machine learning, and artificial intelligence.

Bibliography / References

1. Cisco Networking Academy, Certificate of Completion: “Introduction to Data Science,” issued to Satyam Kumar, completion date 20 June 2026.

2. IILM University, School of Computer Science and Engineering, “Internship Report Format updated 26-27,” supplied report-format document.